

CS352

Computer Graphics

Lecture 10

Two dimensional geometric
transformation

(Composite Transformations)



Matrix Representations and Homogeneous Coordinates

- Many graphics applications involve sequences of geometric transformations.
- Hence we consider how the matrix representations can be reformulated so that such transformation sequence can be efficiently processed.
- Each of three basic two-dimensional transformations (translation, rotation and scaling) can be expressed in the general matrix form

$$\mathbf{P}' = \mathbf{M}_1 \cdot \mathbf{P} + \mathbf{M}_2$$



Matrix Representations and Homogeneous Coordinates

- To produce a sequence of transformations we could calculate the transformed coordinates one step at a time.

Homogeneous Coordinates

- The three geometric transformations can be combined into a single matrix if we expand the representations to 3 by 3 matrices
- Then, we can use the third column of a transformation matrix for the translation terms
- All transformation equations can be expressed as matrix multiplications



Matrix Representations and Homogeneous Coordinates

Homogeneous Coordinates

- A standard technique: expand each two-dimensional coordinate-position representation (x, y) to a three-element representation (x_h, y_h, h) , called homogeneous coordinates
- The homogeneous parameter h is a nonzero value such that
$$x = \frac{x_h}{h}, \quad y = \frac{y_h}{h}$$
- A convenient choice is simply to set $h=1$.
- Each two-dimensional position is then represented with homogeneous coordinate $(x, y, 1)$.



Matrix Representations and Homogeneous Coordinates

Homogeneous Coordinates

- Translation:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{T}(t_x, t_y) \cdot \mathbf{P}$$

Example

For example, to translate a triangle with vertices at original coordinates (10,20), (10,10), (20,10) by $t_x=5$, $t_y=10$, we compute as followings:

Translation of vertex (10,20):

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ 1 \end{bmatrix} = \begin{bmatrix} 1*10 + 0*20 + 5*1 \\ 0*10 + 1*20 + 10*1 \\ 0*10 + 0*20 + 1*1 \end{bmatrix} = \begin{bmatrix} 15 \\ 30 \\ 1 \end{bmatrix}$$

Translation of vertex (10,10):

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 1 \end{bmatrix} = \begin{bmatrix} 1*10 + 0*10 + 5*1 \\ 0*10 + 1*10 + 10*1 \\ 0*10 + 0*10 + 1*1 \end{bmatrix} = \begin{bmatrix} 15 \\ 20 \\ 1 \end{bmatrix}$$

Translation of vertex (20,10):

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 20 \\ 10 \\ 1 \end{bmatrix} = \begin{bmatrix} 1*20 + 0*10 + 5*1 \\ 0*20 + 1*10 + 10*1 \\ 0*20 + 0*10 + 1*1 \end{bmatrix} = \begin{bmatrix} 25 \\ 20 \\ 1 \end{bmatrix}$$

The resultant coordinates of the triangle vertices are (15,30), (15,20), and (25,20) respectively.



Matrix Representations and Homogeneous Coordinates

Homogeneous Coordinates

- Rotation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{R}(\theta) \cdot \mathbf{P}$$

Example: Rotate a polygonal object defined by vertices A(0,0), B(1,0), C(1,1) and D(0,1) by 45° about the origin

$$v = [A \quad B \quad C \quad D] = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$T_{Rotate} = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$v' = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{2}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & 1 & 1 & 1 \end{bmatrix}$$



Example

For example, to rotate a triangle about the origin with vertices at original coordinates (10,20), (10,10), (20,10) by 30 degrees, we compute as followings:

$$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos 30 & -\sin 30 & 0 \\ \sin 30 & \cos 30 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation of vertex (10,20):

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.866 * 10 + (-0.5) * 20 + 0 * 1 \\ 0.5 * 10 + 0.866 * 20 + 0 * 1 \\ 0 * 10 + 0 * 20 + 1 * 1 \end{bmatrix} = \begin{bmatrix} -1.34 \\ 22.32 \\ 1 \end{bmatrix}$$



Example

Rotation of vertex (10,10):

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.866 * 10 + (-0.5) * 10 + 0 * 1 \\ 0.5 * 10 + 0.866 * 10 + 0 * 1 \\ 0 * 10 + 0 * 10 + 1 * 1 \end{bmatrix} = \begin{bmatrix} 3.66 \\ 13.66 \\ 1 \end{bmatrix}$$

Rotation of vertex (20,10):

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 20 \\ 10 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.866 * 20 + (-0.5) * 10 + 0 * 1 \\ 0.5 * 20 + 0.866 * 10 + 0 * 1 \\ 0 * 20 + 0 * 10 + 1 * 1 \end{bmatrix} = \begin{bmatrix} 12.32 \\ 18.66 \\ 1 \end{bmatrix}$$

The resultant coordinates of the triangle vertices are (-1.34,22.32), (3.6,13.66), and (12.32,18.66) respectively.



Matrix Representations and Homogeneous Coordinates

Homogeneous Coordinates

- Scaling:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{S}(s_x, s_y) \cdot \mathbf{P}$$



Example

For example, to scale a triangle with respect to the origin, with vertices at original coordinates (10,20), (10,10), (20,10) by $s_x=2$, $s_y=1.5$, we compute as followings:

Scaling of vertex (10,20):

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 * 10 + 0 * 20 + 0 * 1 \\ 0 * 10 + 1.5 * 20 + 0 * 1 \\ 0 * 10 + 0 * 20 + 1 * 1 \end{bmatrix} = \begin{bmatrix} 20 \\ 30 \\ 1 \end{bmatrix}$$

Scaling of vertex (10,10):

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 * 10 + 0 * 10 + 0 * 1 \\ 0 * 10 + 1.5 * 10 + 0 * 1 \\ 0 * 10 + 0 * 10 + 1 * 1 \end{bmatrix} = \begin{bmatrix} 20 \\ 15 \\ 1 \end{bmatrix}$$

Scaling of vertex (20,10):

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 20 \\ 10 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 * 20 + 0 * 10 + 0 * 1 \\ 0 * 20 + 1.5 * 10 + 0 * 1 \\ 0 * 20 + 0 * 10 + 1 * 1 \end{bmatrix} = \begin{bmatrix} 40 \\ 15 \\ 1 \end{bmatrix}$$

The resultant coordinates of the triangle vertices are (20,30), (20,15), and (40,15) respectively.

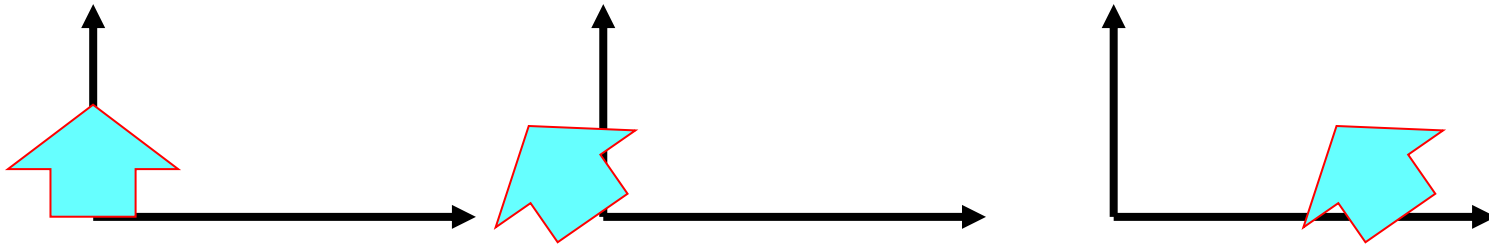
Composite Transformations

Transformation T followed by
Transformation Q followed by
Transformation R:

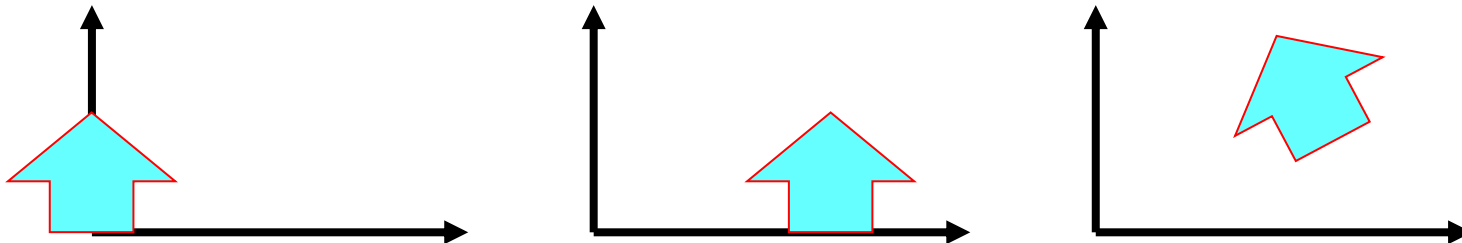
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = [R][Q][T] \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

In composite transformations, the order of transformations is very important.

Rotation followed by Translation:



Translation followed by Rotation:



Combining transformations

$$\mathbf{P}' = \mathbf{M}_1 \mathbf{P} \quad \text{first transformation...}$$

$$\mathbf{P}'' = \mathbf{M}_2 \mathbf{P}' \quad \text{second transformation...}$$

Combined :

$$\begin{aligned} \mathbf{P}'' &= \mathbf{M}_2 (\mathbf{M}_1 \mathbf{P}) \\ &= \mathbf{M}_2 \mathbf{M}_1 \mathbf{P} \\ &= \mathbf{M} \mathbf{P} \quad \text{with } \mathbf{M} = \mathbf{M}_2 \mathbf{M}_1 \end{aligned}$$



Composite Transformations

(two successive translations)

- Composite 2D Translation $T = \mathbf{T}(t_{x1}, t_{y1}) \cdot \mathbf{T}(t_{x2}, t_{y2})$
 $= \mathbf{T}(t_{x1} + t_{x2}, t_{y1} + t_{y2})$

$$\begin{bmatrix} 1 & 0 & t_{x1} \\ 0 & 1 & t_{y1} \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & t_{x2} \\ 0 & 1 & t_{y2} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1*1+0*0+t_{x1}*0 & 1*0+0*1+t_{x1}*0 & 1*t_{x2}+0*t_{y2}+t_{x1}*1 \\ 0*1+1*0+t_{y1}*0 & 0*0+1*1+t_{y1}*0 & 0*t_{x2}+1*t_{y2}+t_{y1}*1 \\ 0*1+0*0+1*0 & 0*0+0*1+1*0 & 0*t_{x2}+0*t_{y2}+1*1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & t_{x1} + t_{x2} \\ 0 & 1 & t_{y1} + t_{y2} \\ 0 & 0 & 1 \end{bmatrix}$$

This demonstrates that 2 successive translations are additive.



Composite Transformations

(two successive scaling's)

- Composite 2D Scaling $T = S(s_{x1}, s_{y1}) \cdot S(s_{x2}, s_{y2})$
 $= S(s_{x1}s_{x2}, s_{y1}s_{y2})$

$$\begin{bmatrix} s_{x1} & 0 & 0 \\ 0 & s_{y1} & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} s_{x2} & 0 & 0 \\ 0 & s_{y2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} s_{x1} * s_{x2} + 0 * 0 + 0 * 0 & s_{x1} * 0 + 0 * s_{y2} + 0 * 0 & s_{x1} * 0 + 0 * 0 + 0 * 1 \\ 0 * s_{x2} + s_{y1} * 0 + 0 * 0 & 0 * 0 + s_{y1} * s_{y2} + 0 * 0 & 0 * 0 + s_{y1} * 0 + 0 * 1 \\ 0 * s_{x2} + 0 * 0 + 1 * 0 & 0 * 0 + 0 * s_{y2} + 1 * 0 & 0 * 0 + 0 * 0 + 1 * 1 \end{bmatrix}$$

$$= \begin{bmatrix} s_{x1} * s_{x2} & 0 & 0 \\ 0 & s_{y1} * s_{y2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This demonstrates that 2 successive scalings with respect to the origin are multiplicative



Composite Transformations

(two successive rotations)

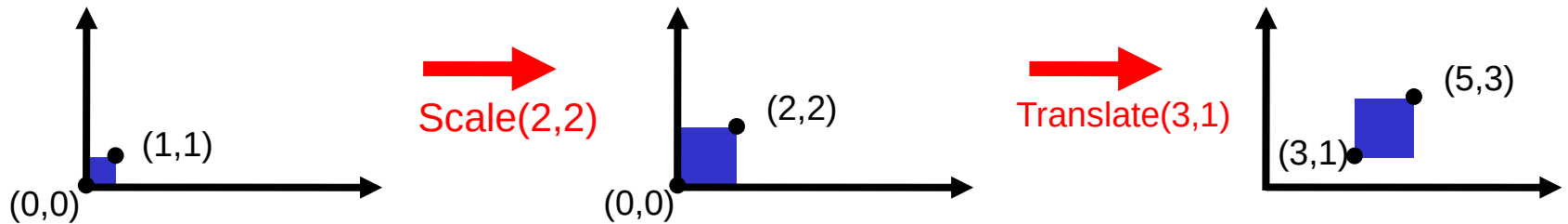
- Composite 2D Rotation $T = \mathbf{R}(\theta) \cdot \mathbf{R}(\alpha)$
 $= \mathbf{R}(\theta + \alpha)$

$$\begin{aligned} & \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta \cos \alpha + (-\sin \theta) * \sin \alpha + 0 * 0 & \cos \theta * (-\sin \alpha) + (-\sin \theta) * \cos \alpha + 0 * 0 & \cos \theta * 0 + (-\sin \theta) * 0 + 0 * 1 \\ \sin \theta \cos \alpha + \cos \theta * \sin \alpha + 0 * 0 & \sin \theta * (-\sin \alpha) + \cos \theta * \cos \alpha + 0 * 0 & \sin \theta * 0 + \cos \theta * 0 + 0 * 1 \\ 0 * \cos \alpha + 0 * \sin \alpha + 1 * 0 & 0 * (-\sin \alpha) + 0 * \cos \alpha + 1 * 0 & 0 * 0 + 0 * 0 + 1 * 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta \cos \alpha - \sin \theta \sin \alpha & -(\cos \theta \sin \alpha + \sin \theta \cos \alpha) & 0 \\ \sin \theta \cos \alpha + \cos \theta \sin \alpha & -\sin \theta \sin \alpha + \cos \theta \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos(\theta + \alpha) & -\sin(\theta + \alpha) & 0 \\ \sin(\theta + \alpha) & \cos(\theta + \alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

This demonstrates that 2 successive rotations are additive

How are transforms combined?

Example 1: Scale then Translate



Use matrix multiplication: $p' = T (S p) = TS p$

$$TS = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Caution: matrix multiplication is NOT commutative!

How are transforms combined?

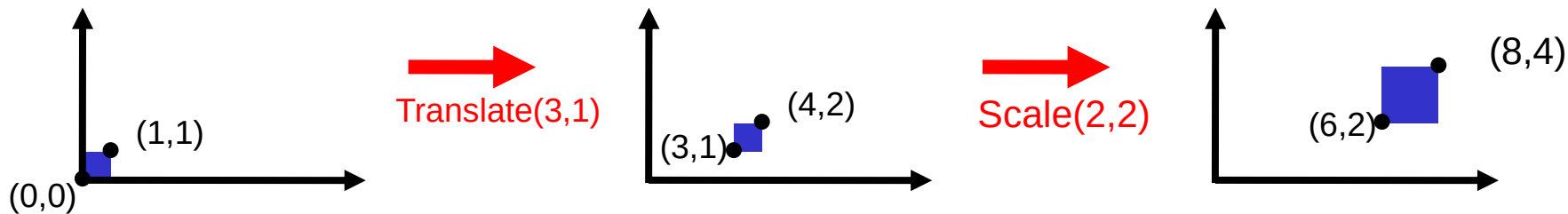
Example 1: Scale then Translate: $p' = T(S p) = TS p$

$$TS = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$TSP = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$$

How are transforms combined?

Example 2: Translate then Scale: $p' = S(T p) = ST p$



Translate then Scale: $p' = S(T p) = ST p$

$$ST = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 6 \\ 0 & 2 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

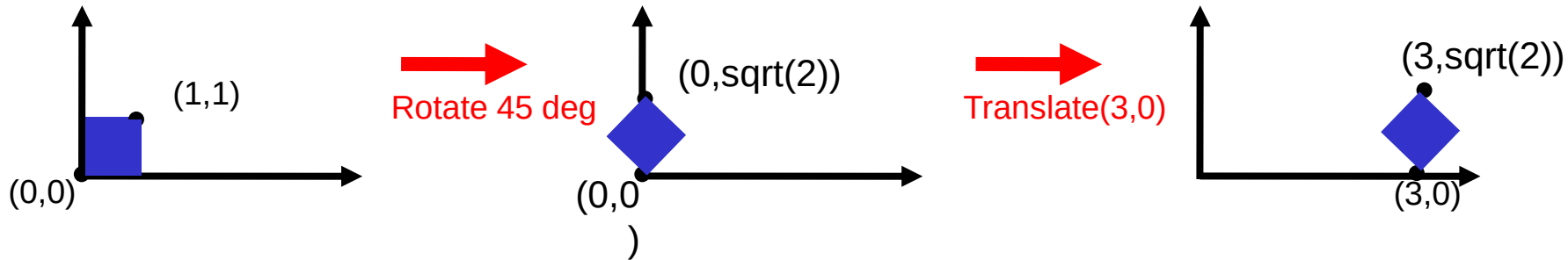
How are transforms combined?

Example 2: Translate then Scale: $p' = S (T p) = ST p$

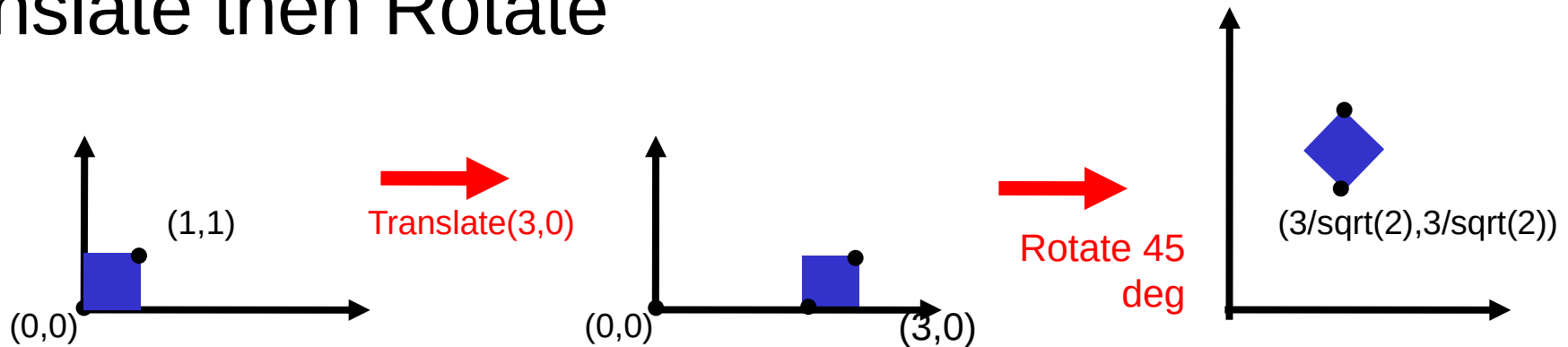
$$STP = \begin{bmatrix} 2 & 0 & 6 \\ 0 & 2 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 2 & 4 \\ 1 & 1 \end{bmatrix}$$

How are transforms combined?

Rotate then Translate



Translate then Rotate



Caution: matrix multiplication is NOT commutative!

Non-commutative Composition

Rotate then Translate: $p' = T (R p) = TR p$

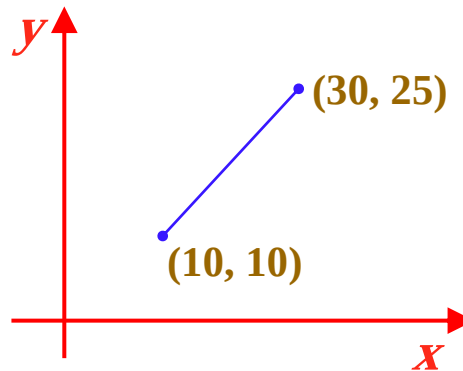
$$TR = \begin{bmatrix} 1 & 0 & \textcolor{red}{3} \\ 0 & 1 & \textcolor{red}{1} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \textcolor{red}{0} & -1 & \textcolor{red}{3} \\ 0 & \textcolor{red}{0} & \textcolor{red}{1} \\ 0 & 0 & 1 \end{bmatrix}$$

Translate then Rotate: $p' = R (T p) = RT p$

$$RT = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \textcolor{red}{3} \\ 0 & 1 & \textcolor{red}{1} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & \textcolor{red}{-1} \\ 1 & \textcolor{red}{0} & \textcolor{red}{3} \\ 0 & 0 & \textcolor{red}{1} \end{bmatrix}$$

Example 3

- Consider the line with endpoints $(10, 10)$ and $(30, 25)$. Translate it by $t_x = -20$, $t_y = -10$ and then rotate it by $\theta = 90^\circ$.



Right-to-left

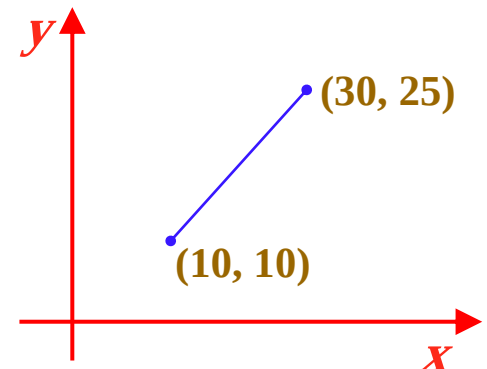
$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} \cos 90^\circ & -\sin 90^\circ & 0 \\ \sin 90^\circ & \cos 90^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -20 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

Solution

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} \cos 90^\circ & -\sin 90^\circ & 0 \\ \sin 90^\circ & \cos 90^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -20 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

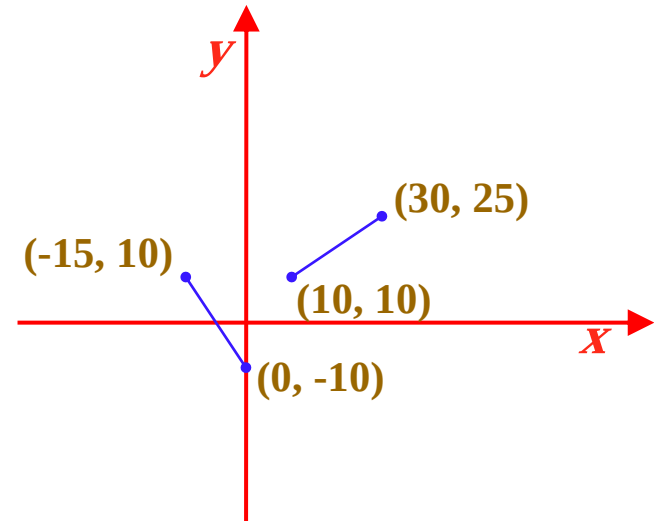
$$= \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -20 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -1 & 10 \\ 1 & 0 & -20 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$



Solution (continue)

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 10 \\ 1 & 0 & -20 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$



Point (10, 10)

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 10 \\ 1 & 0 & -20 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 10 \\ 10 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -10 \\ 1 \end{pmatrix}$$

Point (30, 25)

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 10 \\ 1 & 0 & -20 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 30 \\ 25 \\ 1 \end{pmatrix} = \begin{pmatrix} -15 \\ 10 \\ 1 \end{pmatrix}$$

Inverse Transformations

- Inverse translation matrix:

$$\mathbf{T}^{-1}(t_x, t_y) = \mathbf{T}(-t_x, -t_y) \quad \longrightarrow \quad \mathbf{T}^{-1} = \begin{bmatrix} 1 & 0 & -t_x \\ 0 & 1 & -t_y \\ 0 & 0 & 1 \end{bmatrix}$$

- Inverse rotation matrix:

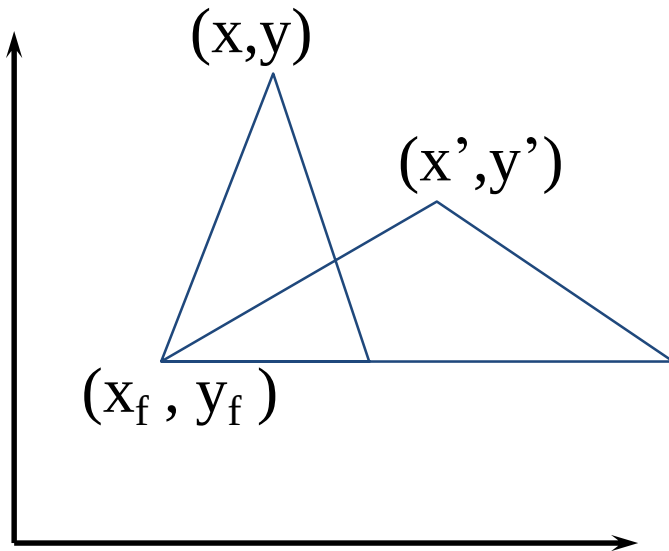
$$\mathbf{R}^{-1}(\theta) = \mathbf{R}^T(\theta) \quad \longrightarrow \quad \mathbf{R}^{-1} = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Inverse Transformations

- Inverse scaling matrix:

$$\mathbf{S}^{-1}(s_x, s_y, 1) = \mathbf{S}\left(\frac{1}{s_x}, \frac{1}{s_y}, 1\right) \Rightarrow \mathbf{S}^{-1} = \begin{bmatrix} \frac{1}{s_x} & 0 & 0 \\ 0 & \frac{1}{s_y} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Scaling About a Fixed Point



- Translate the fixed point to origin
- Scale with respect to the origin
- Translate the fixed point to its original position.

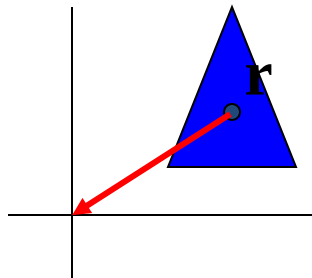
$$x' = (x - x_f) \cdot s_x + x_f$$

$$y' = (y - y_f) \cdot s_y + y_f$$

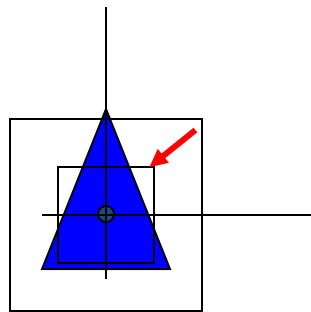
Scaling w.r.t. point

Scale with factors s_x and s_y w.r.t. point r :

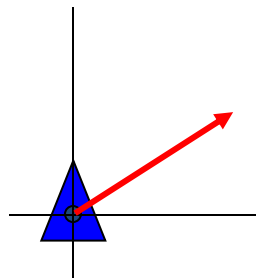
- 1) Translate such that r coincides with origin;
- 2) Scale w.r.t. origin;
- 3) Translate back again.



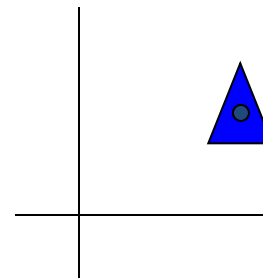
1)



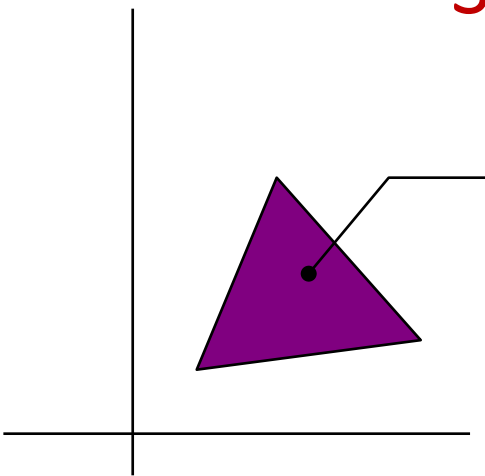
2)



3)



Scaling about a Reference Point



$(x_c, y_c)^T$

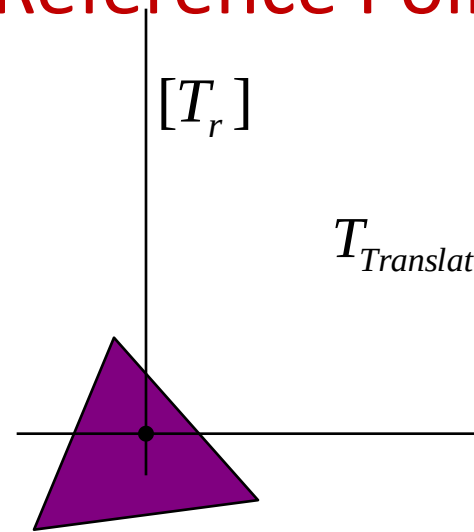
Scale w.r.t. point r :

$$1) \mathbf{P}' = \mathbf{T}(-x_r, -y_r) \mathbf{P}$$

$$2) \mathbf{P}'' = \mathbf{S}(s_x, s_y) \mathbf{P}'$$

$$3) \mathbf{P}''' = \mathbf{T}(x_r, y_r) \mathbf{P}''$$

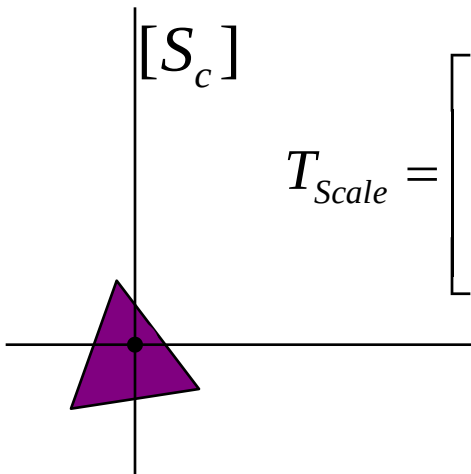
(a) Original position of object and fixed point



$[T_r]$

$$T_{Translate} = \begin{bmatrix} 1 & 0 & -x_r \\ 0 & 1 & -y_r \\ 0 & 0 & 1 \end{bmatrix}$$

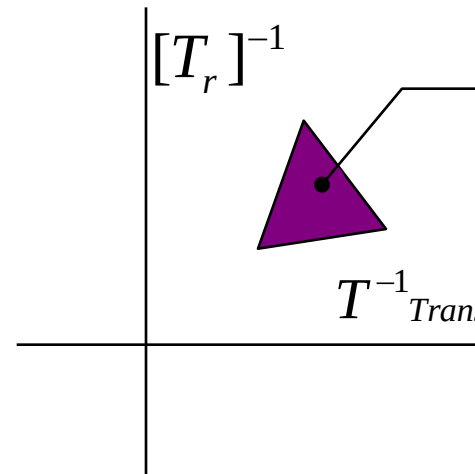
(b) Translate object so that fixed point (x_r, y_r) is at origin



$[S_c]$

$$T_{Scale} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(c) Scale object with respect to origin



$[T_r]^{-1}$

$(x_c, y_c)^T$

$$T_{Translate}^{-1} = \begin{bmatrix} 1 & 0 & x_r \\ 0 & 1 & y_r \\ 0 & 0 & 1 \end{bmatrix}$$

(d) Translate object so that fixed point is returned to position $(x_r, y_r)^T$

Combining the Transformation

The three transformation matrices are combined as follows

$$T_{composite} = T_{trans} T_{Scale} T_{Trans}^{-1}$$

$$T_{Composite} = \begin{bmatrix} 1 & 0 & -x_r \\ 0 & 1 & -y_r \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & x_r \\ 0 & 1 & y_r \\ 0 & 0 & 1 \end{bmatrix}$$

$$T_{Composite} = \begin{bmatrix} S_x & 0 & x_r(S_x - 1) \\ 0 & S_y & y_r(S_y - 1) \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = [T_{composite}] \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Example1:

Consider a triangle having a vertices at A(0,0), B(5,1), and C(3,4) scale this triangle using scaling factors $S_x=2$ and $S_y=1$ about a) the origin b) about the reference point (2,3)

a) At origin ($x_r=0$, $y_r=0$)

$$T_{Composite} = \begin{bmatrix} S_x & 0 & x_r(S_x - 1) \\ 0 & S_y & y_r(S_y - 1) \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$v = [A \quad B \quad C] = \begin{bmatrix} 0 & 5 & 3 \\ 0 & 1 & 4 \\ 1 & 1 & 1 \end{bmatrix}$$

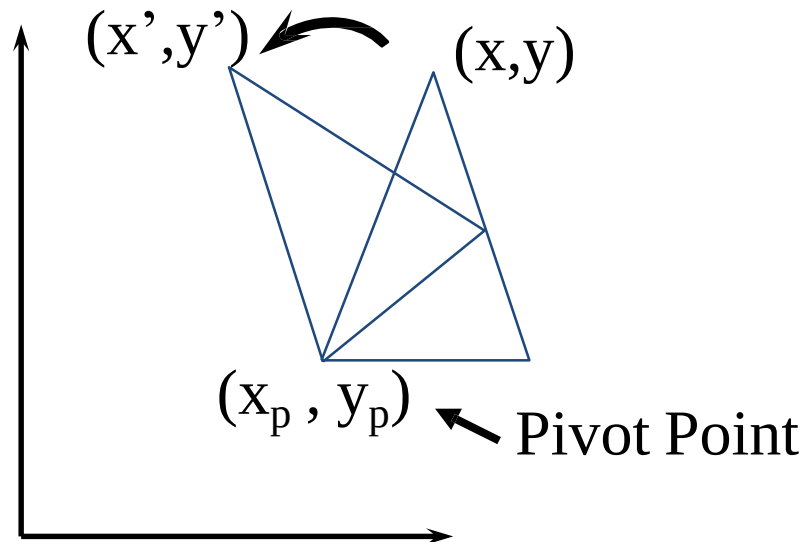
$$v' = T_{composite} * v = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 0 & 5 & 3 \\ 0 & 1 & 4 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 10 & 6 \\ 0 & 0 & 4 \\ 1 & 1 & 1 \end{bmatrix}$$

b) Scaling about the reference point (2,3) : Here, the transformation matrix

$$T_{Composite} = \begin{bmatrix} S_x & 0 & x_r(S_x - 1) \\ 0 & S_y & y_r(S_y - 1) \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$v' = T_{composite} * v = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 0 & 5 & 3 \\ 0 & 1 & 4 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 12 & 8 \\ 0 & 1 & 4 \\ 1 & 1 & 1 \end{bmatrix}$$

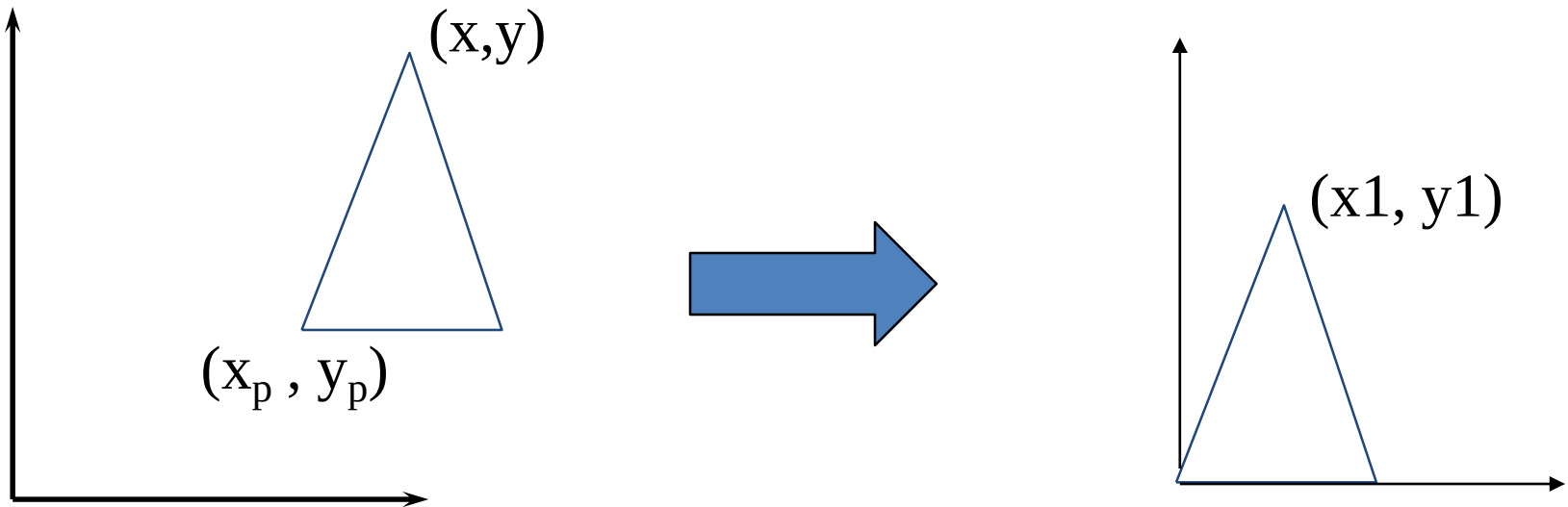
Rotation About a Pivot Point



- Pivot point is the point of rotation
- Pivot point need not necessarily be on the object

Rotation About a Pivot Point

STEP-1: Translate the pivot point to the origin

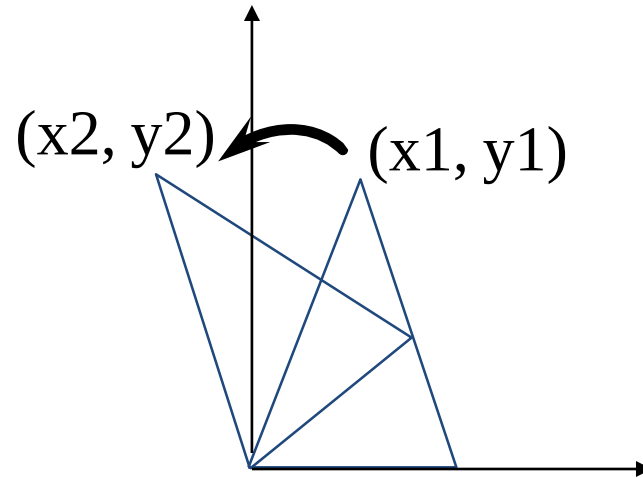


$$x1 = x - x_p$$

$$y1 = y - y_p$$

Rotation About a Pivot Point

STEP-2: Rotate about the origin



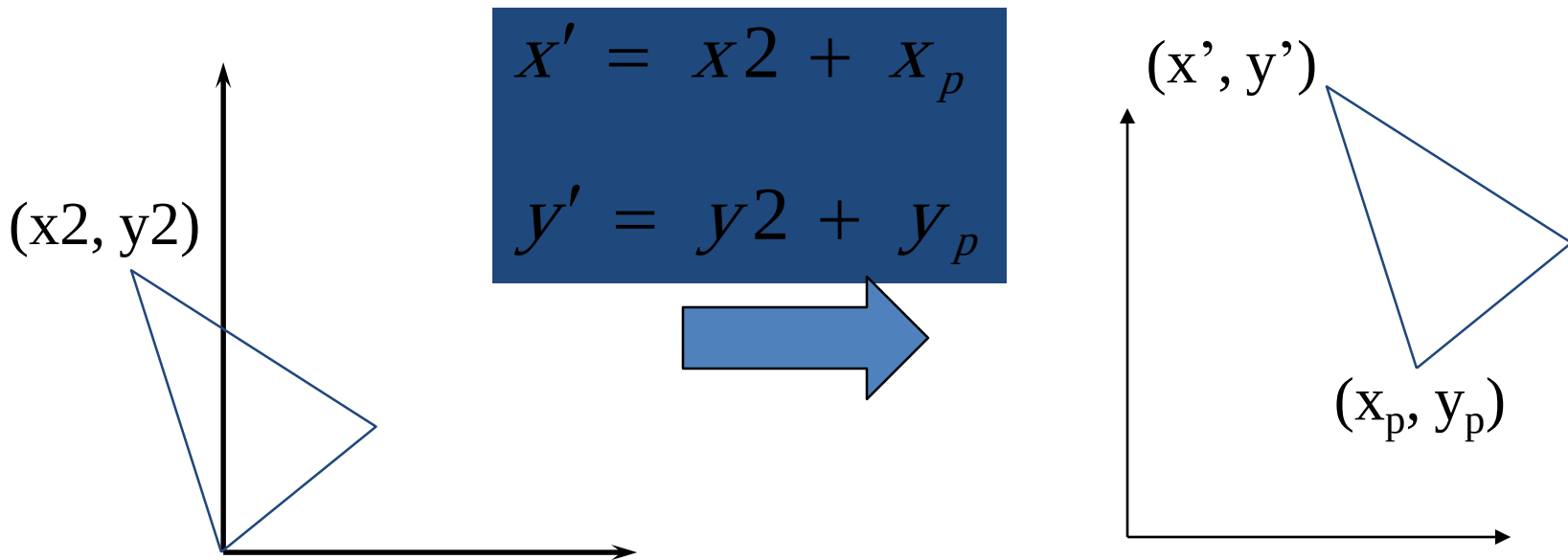
$$x_2 = x_1 \cos \theta - y_1 \sin \theta$$

$$y_2 = x_1 \sin \theta +$$

$$y_1 \cos \theta$$

Rotation About a Pivot Point

STEP-3: Translate the pivot point to original position

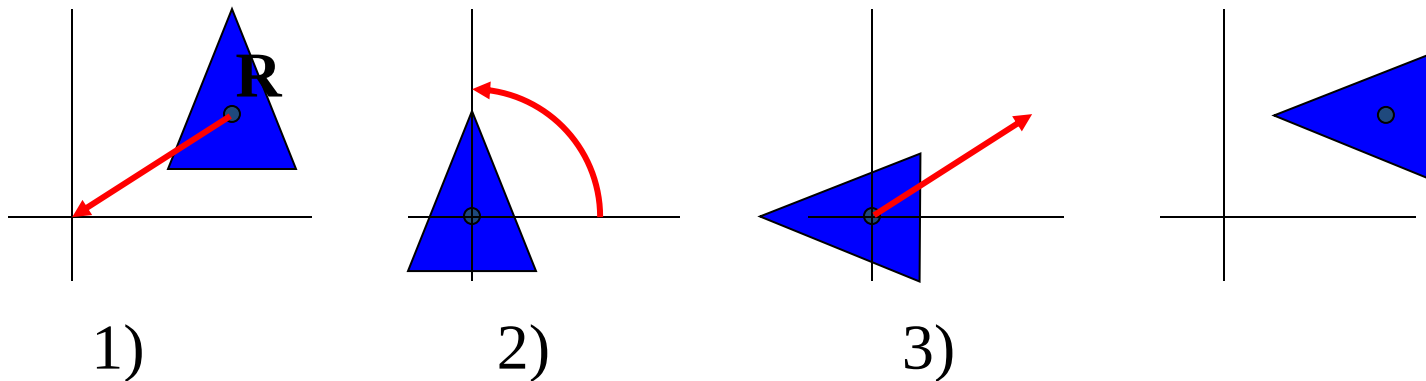


$$\begin{aligned}x' &= (x - x_p) \cos \theta - (y - y_p) \sin \theta + x_p \\ y' &= (x - x_p) \sin \theta + (y - y_p) \cos \theta + y_p\end{aligned}$$

Rotation around a point

Rotate over angle θ around point **R** :

- 1) Translate such that **R** coincides with origin;
- 2) Rotate over angle θ around origin;
- 3) Translate back.



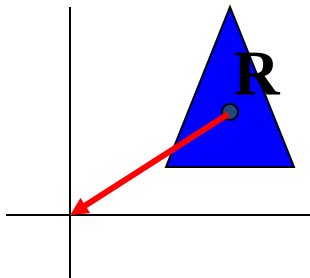
Rotation around a point

Rotate over angle θ around point **R** :

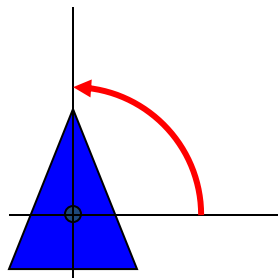
$$1) \mathbf{P}' = \mathbf{T}(-R_x, -R_y) \mathbf{P}$$

$$2) \mathbf{P}'' = \mathbf{R}(\theta) \mathbf{P}'$$

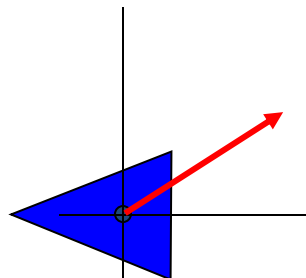
$$3) \mathbf{P}''' = \mathbf{T}(R_x, R_y) \mathbf{P}''$$



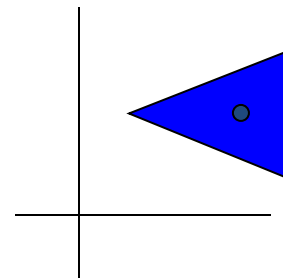
1)



2)



3)

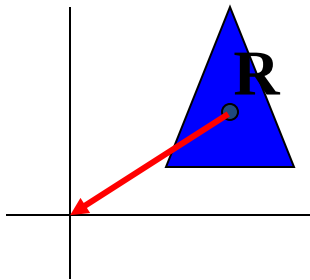


Rotation around point

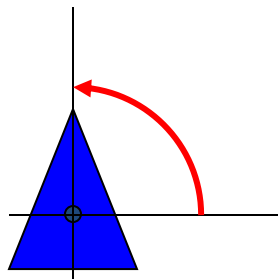
$$1) \mathbf{P}' = \mathbf{T}(-R_x, -R_y) \mathbf{P}$$

$$2) \mathbf{P}'' = \mathbf{R}(\theta) \mathbf{P}' = \mathbf{R}(\theta) \mathbf{T}(-R_x, -R_y) \mathbf{P}$$

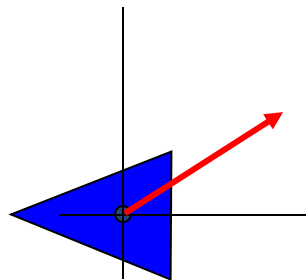
$$\begin{aligned} 3) \mathbf{P}''' &= \mathbf{T}(R_x, R_y) \mathbf{P}'' \\ &= \mathbf{T}(R_x, R_y) \mathbf{R}(\theta) \mathbf{P}' \\ &= \mathbf{T}(R_x, R_y) \mathbf{R}(\theta) \mathbf{T}(-R_x, -R_y) \mathbf{P} \end{aligned}$$



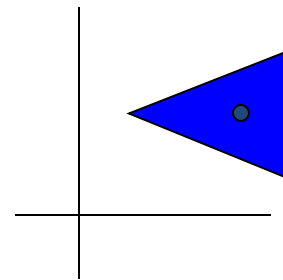
1)



2)



3)

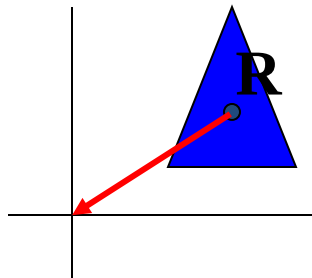


Rotation around point

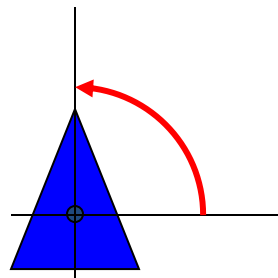
$$\mathbf{P}''' = \mathbf{T}(R_x, R_y) \mathbf{R}(\theta) \mathbf{T}(-R_x, -R_y) \mathbf{P}$$

or

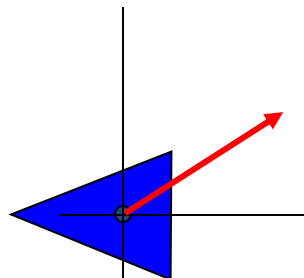
$$\mathbf{P}''' = \begin{pmatrix} \cos \theta & -\sin \theta & R_x(1 - \cos \theta) + R_y \sin \theta \\ \sin \theta & \cos \theta & R_y(1 - \cos \theta) - R_x \sin \theta \\ 0 & 0 & 1 \end{pmatrix} \mathbf{P}$$



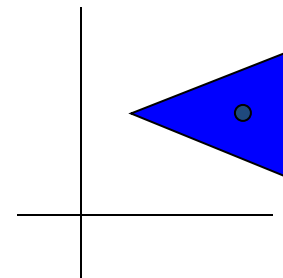
1)



2)



3)



Rotation About a Pivot Point

$$x' = (x - x_p) \cos \theta - (y - y_p) \sin \theta + x_p$$
$$y' = (x - x_p) \sin \theta + (y - y_p) \cos \theta + y_p$$

Specifying a 2D-Rotation about a pivot point (xp,yp):

```
glTranslatef(xp, yp, 0);  
glRotatef(theta, 0, 0, 1.0);  
glTranslatef(-xp, -yp, 0);
```

Note the OpenGL specification of the sequence of transformations in the reverse order !

Example: Consider a triangle having vertices at A(0,0), B(5,1), and C(3,4). Rotate the triangle by 90° about a) the origin

b) about the reference point (2,3)

Solution: a) about the origin

$$v = [A \quad B \quad C] = \begin{bmatrix} 0 & 5 & 3 \\ 0 & 1 & 4 \\ 1 & 1 & 1 \end{bmatrix}$$

The transformation matrix for rotation about the origin :

$$T_{Rotate} = \begin{bmatrix} \cos 90^\circ & -\sin 90^\circ & 0 \\ \sin 90^\circ & \cos 90^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$v' = T_{Rot} \bullet v$$

$$v' = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 5 & 3 \\ 0 & 1 & 4 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & -4 \\ 0 & 5 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

b) Rotation about point (2,3)

$$T_{composite} = T_{trans} T_{Rot} T_{Trans}^{-1}$$

$$T_{composite} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T_{composite} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & -3 \\ 1 & 0 & -2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & -5 \\ 1 & 0 & -5 \\ 0 & 0 & 1 \end{bmatrix}$$

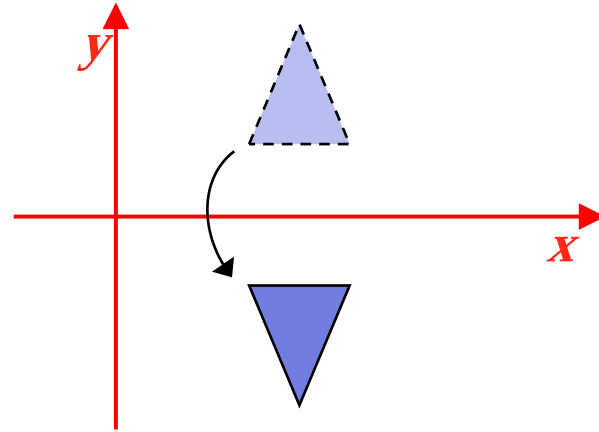
$$v' = T_{Composite} \bullet v$$

$$v' = \begin{bmatrix} 0 & -1 & -5 \\ 1 & 0 & -5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 5 & 3 \\ 0 & 1 & 4 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -5 & -6 & -9 \\ -5 & 0 & -2 \\ 1 & 1 & 1 \end{bmatrix}$$

Reflection

About the x axis

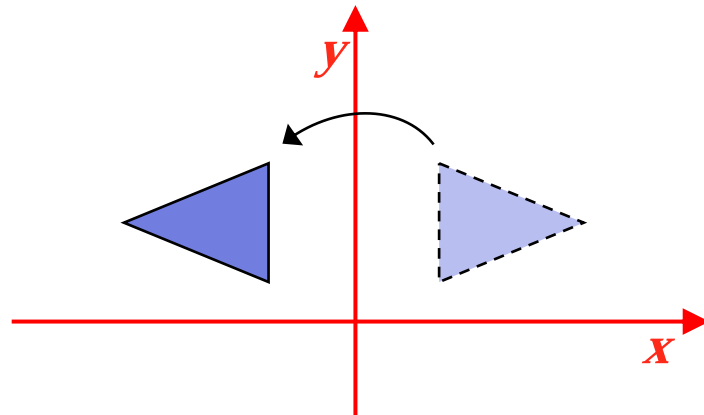
$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$



Reflection of an object about the x axis

About the y axis

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

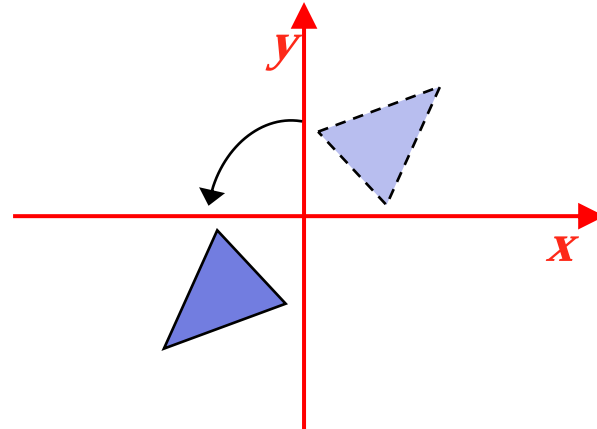


Reflection of an object about the y axis

Reflection

Relative to the coordinate origin

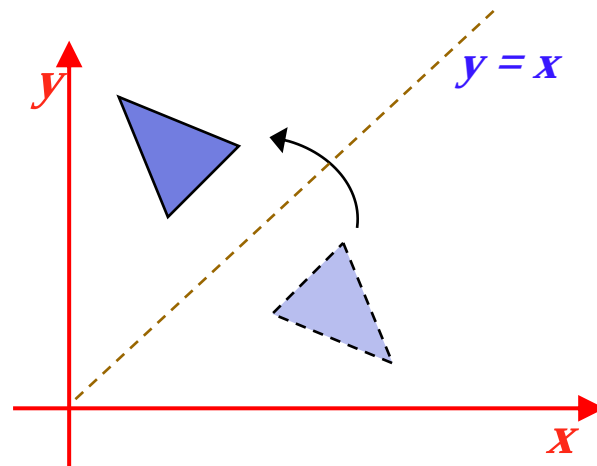
$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$



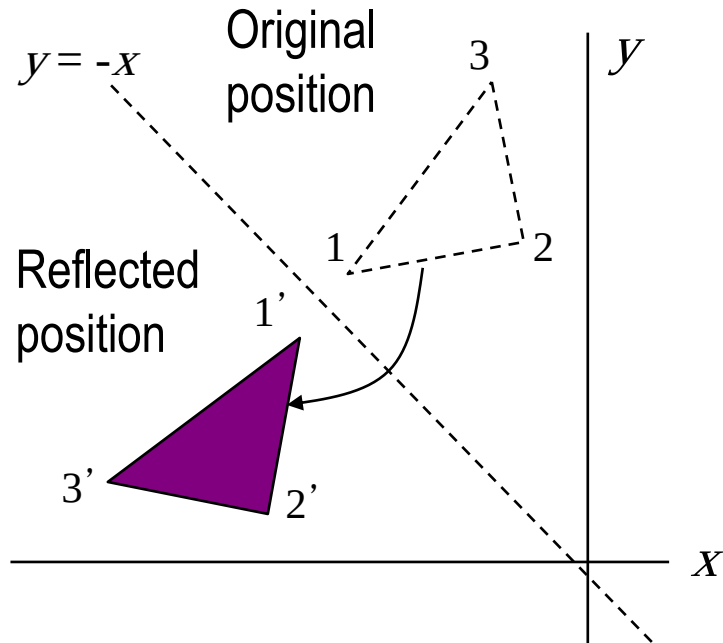
Same as a rotation with 180°

With respect to the line $y = x$

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$



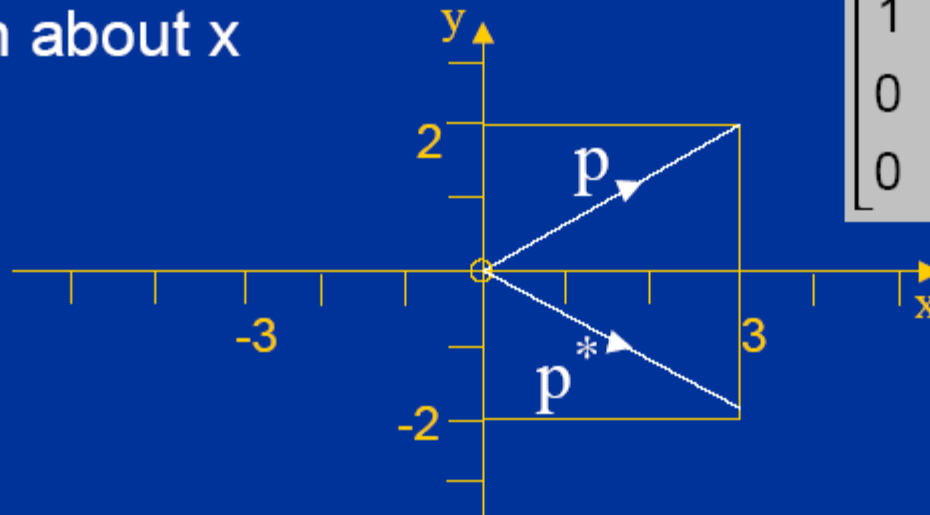
Reflection about $x=-y$ line : the x and y coordinates are interchanged and their values are negated after reflection



$$T_{refx=-y} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

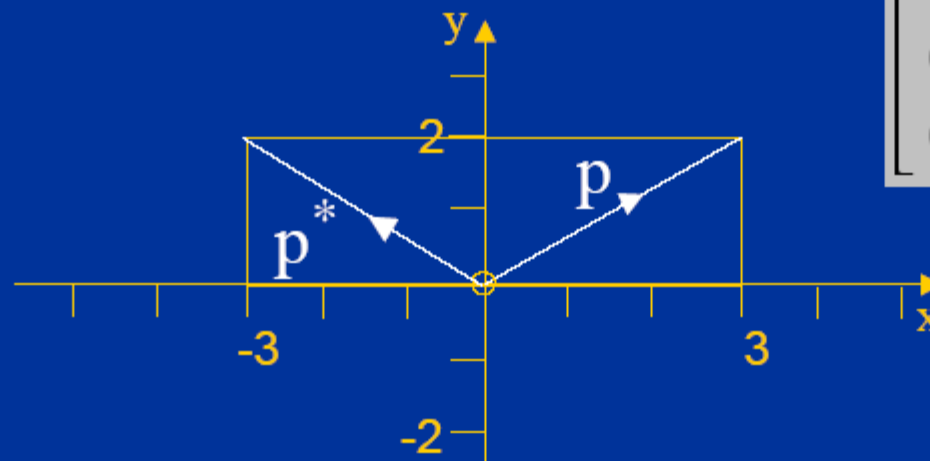
Example: Reflection of Points

Reflection about x



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$$

Reflection about y

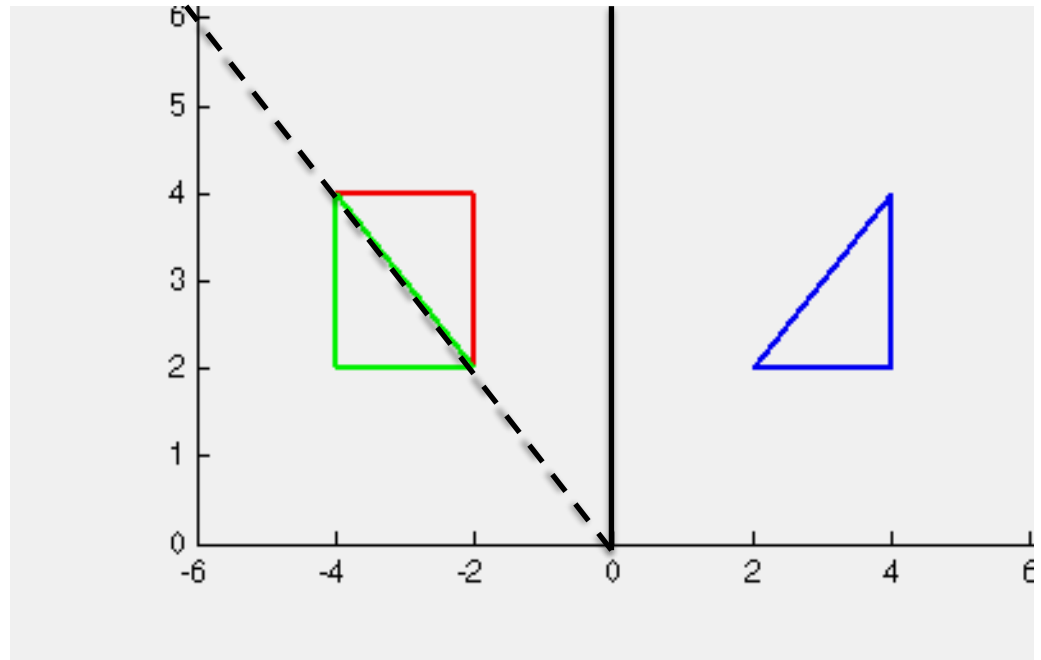


$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}$$

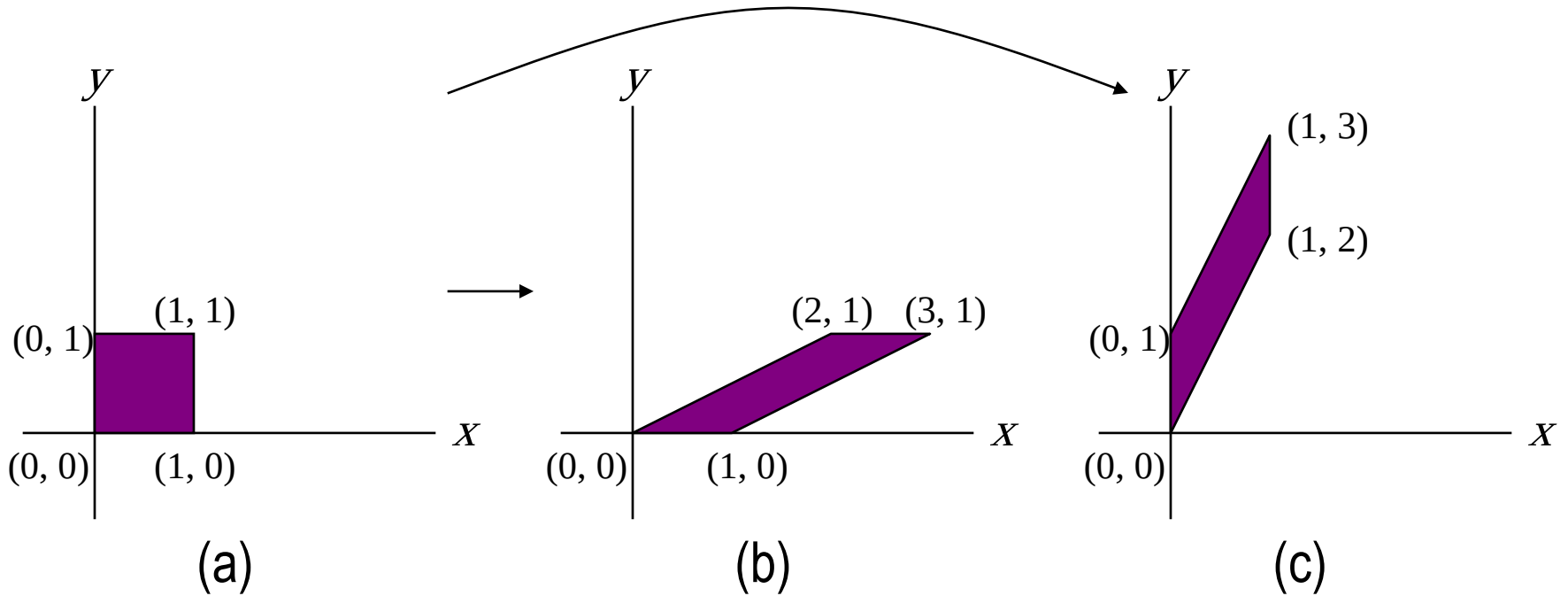
Example: Consider a triangle whose vertices are (2 2), (4 2) and (4 4). Find the concatenated transformation matrix and the transformed vertices for rotation of 90 about the origin followed by reflection through the line $y = -x$. Comment on the sequence of transformations.

$$[T_{rot90}] = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[T_{ref\ y=-x}] = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Shearing about the origin



$$SH_x = 2$$

$$T_{shearx} = \begin{bmatrix} 1 & SH_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$SH_y = 2$$

$$T_{sheary} = \begin{bmatrix} 1 & 0 & 0 \\ SH_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Shearing about the a reference point (x_r, y_r)

$$T_{shearx} = \begin{bmatrix} 1 & SH_x & SH_x \times y_{ref} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T_{sheary} = \begin{bmatrix} 1 & 0 & 0 \\ SH_y & 1 & SH_y \times x_{ref} \\ 0 & 0 & 1 \end{bmatrix}$$